



Module Presenter's Manual

for

Working with MySQL

Effective from: February, 2025

Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2025	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	2
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	9

1. Introduction

At the end of this course, students will be able to:

- Understand basics of MySQL
- Understand various installation steps for MySQL
- Learn commands to create, view and alter a database and table
- Learn various commands to work with data
- Identify the different types of table joins in MySQL
- Use various functions in MySQL
- Create and use Stored Procedures
- Explain transactions and how to handle them
- Describe Performance Optimization
- Explain capability of MySQL for scaling and availability
- Explain replication in MySQL
- Perform data management using replication
- Understand concepts of partitioning in MySQL
- Understand concepts of Storage Systems and Management

2. Information on Session Allocation

Module	Online Hours	Self Study Hours
Working with MySQL	32	10

Throughout this Presenter's Manual, the module **Working with MySQL** will be referred to as **MySQL**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF.
Download eBook - Student has the option to download the subject related e-book and read offline.
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.
Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
Glossary - Student can access a list of subject related specialized words with their definitions.
FAQ - Student can access frequently asked questions and their answers.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4	
1	Session 1 MySQL – TL1	Session 2 MySQL – TL2	Session 3 MySQL – TL3	Session 4 MySQL – TL4	
2	Session 5 MySQL – TL5	Session 6 MySQL – TL6	Session 7 MySQL – TL7	Session 8 MySQL – TL8	S1-S3
3	Session 9 MySQL – TL9	Session 10 MySQL – TL10	Session 11 MySQL – TL11	Session 12 MySQL – TL12	
4	Session 13 MySQL – TL13	Session 14 MySQL – TL14	Session 15 MySQL – TL15	Session 16 MySQL – TL16	S4-S5

MySQL: Working with MySQL

TL: Online Session

S: Self Study Session (over and above the module duration)

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	MySQL – TL1	All the topics as listed below from Session 1 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 1 - Introduction to MySQL</u> ➤ Define database ➤ List and explain different elements of database management system ➤ Describe the principles of MySQL ➤ List different types of database models ➤ Define RDBMS and its terminology ➤ Establish MySQL Connection ➤ Outline the concepts of Database Design in DBMS ➤ Explain how to connect to Database Server Using MySQL Workbench ➤ Explain features, limitations, and deployment of MySQL ➤ Explain different elements of Normalization in DBMS	Practical MySQL SG-Session 1 XP-Session 1 TG-Session 1
2	MySQL – TL2	All the topics as listed below from Session 2 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 2 – Essentials of MySQL</u> ➤ List MySQL Data Types ➤ List and explain different commands in MySQL ➤ Explain the basics of creating a database ➤ Identify ways to create and drop a database in MySQL ➤ Identify ways to select a database in MySQL ➤ Outline how to create, alter and drop a table	Practical MySQL SG-Session 2 XP-Session 2 TG-Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	MySQL – TL3	All the topics as listed below from Session 3 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 3 – Working with Data in MySQL</u> ➤ Explain the SELECT command in detail ➤ Identify the use of INSERT, UPDATE, and DELETE in the manipulation of data ➤ Outline the use of LIKE, IN, BETWEEN, and DISTINCT clauses ➤ Explain MERGE command and its usage	Practical MySQL SG-Session 3 XP-Session 3 TG-Session 3
4	MySQL – TL4	All the topics as listed below from Session 4 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 4 – Joins</u> ➤ Define JOINS ➤ Explain the use of JOINS ➤ Identify different types of JOINS ➤ Explain the use of SET operators ➤ Outline the differences between JOINS and UNIONS	Practical MySQL SG-Session 4 XP-Session 4 TG-Session 4
5	MySQL – TL5	The Workshops of Session 1 to 4 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL (OnlineVarsity) Session 1-4
6	MySQL – TL6	All the topics as listed below from Session 5 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 5 – Subqueries</u> ➤ Explain subqueries and their usage ➤ Outline the clauses and keywords used in subqueries ➤ Describe practical uses of WHERE and FROM clauses in subqueries ➤ Define IN, NOT IN, EXISTS, and NOT EXISTS keywords ➤ List and explain different types of subqueries	Practical MySQL SG-Session 5 XP-Session 5 TG-Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
7	MySQL – TL7	All the topics as listed below from Session 6 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 6 – Functions and Stored Procedures</u> ➤ Explain the use of functions in MySQL ➤ Explain different types of functions ➤ Describe the usage of stored routines ➤ Explain how functions are created ➤ Outline the uses of stored procedures and functions	Practical MySQL SG-Session 6 XP-Session 6 TG-Session 6
8	MySQL – TL8	All the topics as listed below from Session 7 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 7 – MySQL Clauses and Indexes</u> ➤ Explain the use of functions in Describe MySQL HAVING and ORDER BY clause ➤ Describe MySQL GROUP BY clause ➤ Define and use ROLLUP Modifier ➤ Explain Indexes in MySQL	Practical MySQL SG-Session 7 XP-Session 7 TG-Session 7
	S1-S3	All the Assignments from Session 1 to Session 7 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL Session 1-7
9	MySQL – TL9	The Workshops of Session 5 to 7 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL (OnlineVarsity) Session 5-7
10	MySQL – TL10	All the topics as listed below from Session 8 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 8 – Transactions, Performance Management and Backup</u> ➤ Define transaction ➤ List different commands used in transactions ➤ Describe transaction using JDBC Driver	Practical MySQL SG-Session 8 XP-Session 8 TG-Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Outline different techniques to improve and manage database performance ➤ Explain Replication	
11	MySQL – TL11	All the topics as listed below from Session 9 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 9 – Replication and Scalability</u> ➤ Explain MySQL replication architecture ➤ Develop and build on replication ➤ Describe replication topologies ➤ List replication use cases ➤ Summarize types of replications ➤ Identify advantages and disadvantages of MySQL replication	Practical MySQL SG-Session 9 XP-Session 9 TG-Session 9
12	MySQL – TL12	All the topics as listed below from Session 10 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 10 – Partitioning</u> ➤ Define partitioning and its different types ➤ Describe how partitioning works ➤ List the benefits of partitioning ➤ Explain partitioning maintenance, pruning and selection ➤ Outline the restrictions and limitations of partitioning	Practical MySQL SG-Session 10 XP-Session 10 TG-Session 10
13	MySQL – TL13	The Workshops from Session 8 to Session 10 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL (OnlineVarsity) Session 8-10
14	MySQL – TL14	All the topics as listed below from Session 11 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 11 – Performance Optimization in MySQL</u>	Practical MySQL SG-Session 11 XP-Session 11 TG-Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain optimizing queries with MySQL query optimization guidelines ➤ Outline the process of optimizing workload ➤ Define how to monitor fundamental resources ➤ Describe restructuring queries ➤ Describe optimizing subqueries ➤ Identify scalability problems ➤ Explain queue cache ➤ Describe and explain automating of configuration ➤ Explain specific types of queries	
15	MYSQL – TL15	All the topics as listed below from Session 12 of <i>Practical MySQL</i> book should be covered in this session. <u>Session 12 – Concept of Storage System and Management</u> ➤ Outline the types of data storage ➤ Explain different types of storage ➤ Describe demand paging and thrashing ➤ Explain process management in DBMS ➤ Define paging and segmentation	Practical MySQL SG-Session 12 XP-Session 12 TG-Session 12
16	MYSQL – TL16	The Workshops from Session 8 to Session 12 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL (OnlineVarsity) Session 9-12
	S4-S5	All the Assignments from Session 8 to Session 12 of <i>Practical MySQL</i> course should be covered in this session.	Practical MySQL Session 8-12

6. Library References

- | |
|--|
| ➤ Learning MySQL by Sergey Kuzmichev and Vinicius M. Grippa |
| ➤ Murach's MySQL by Joel Murach |
| ➤ MySQL(TM): The Complete Reference by Vikram Vaswani |